

CLUSTERING ANALYSIS

Customer Churn Segmentation for a Telecommunications Company

K-Means clustering — 6,741 customers — 4 behavioural segments

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Objective & Approach

What this analysis set out to do



Select & apply K-Means clustering

Group customers by behaviour and demographics, not by whether they already churned.



Determine optimal clusters via the Elbow Method

Cross-checked with Silhouette Score for a statistically defensible choice of k.



Visualise and label clusters

Turn raw cluster numbers into interpretable, business-ready customer segments.

6,741

customers analysed

4

customer segments identified

12

features used in clustering

Methodology: Features & Algorithm

K-Means clustering was applied to 12 standardised customer features, grouped into four categories. The Churn label was deliberately excluded from clustering — kept aside only to interpret the resulting segments.



Demographics

- gender
- SeniorCitizen
- Dependents



Tenure

- tenure (standardised)



Services

- PhoneService
- MultipleLines
- InternetService (DSL / Fiber)

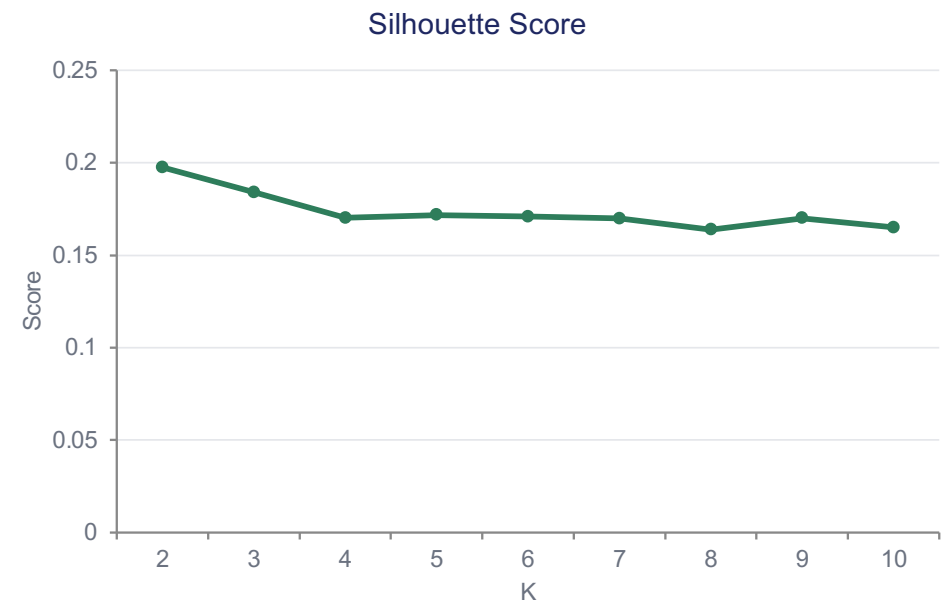
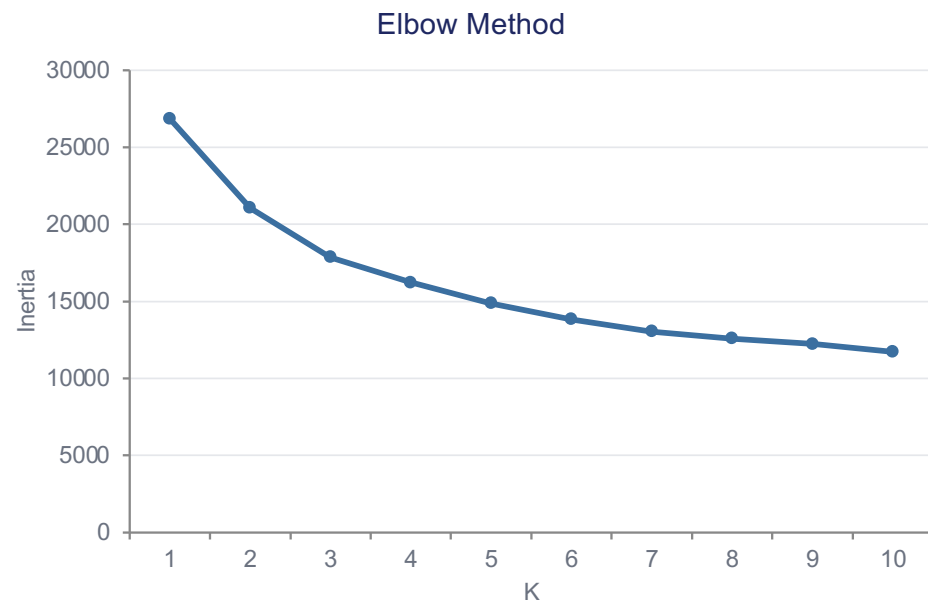


Billing & Contract

- MonthlyCharges
- Contract (Month-to-month / One year / Two year)

Determining the Optimal Number of Clusters

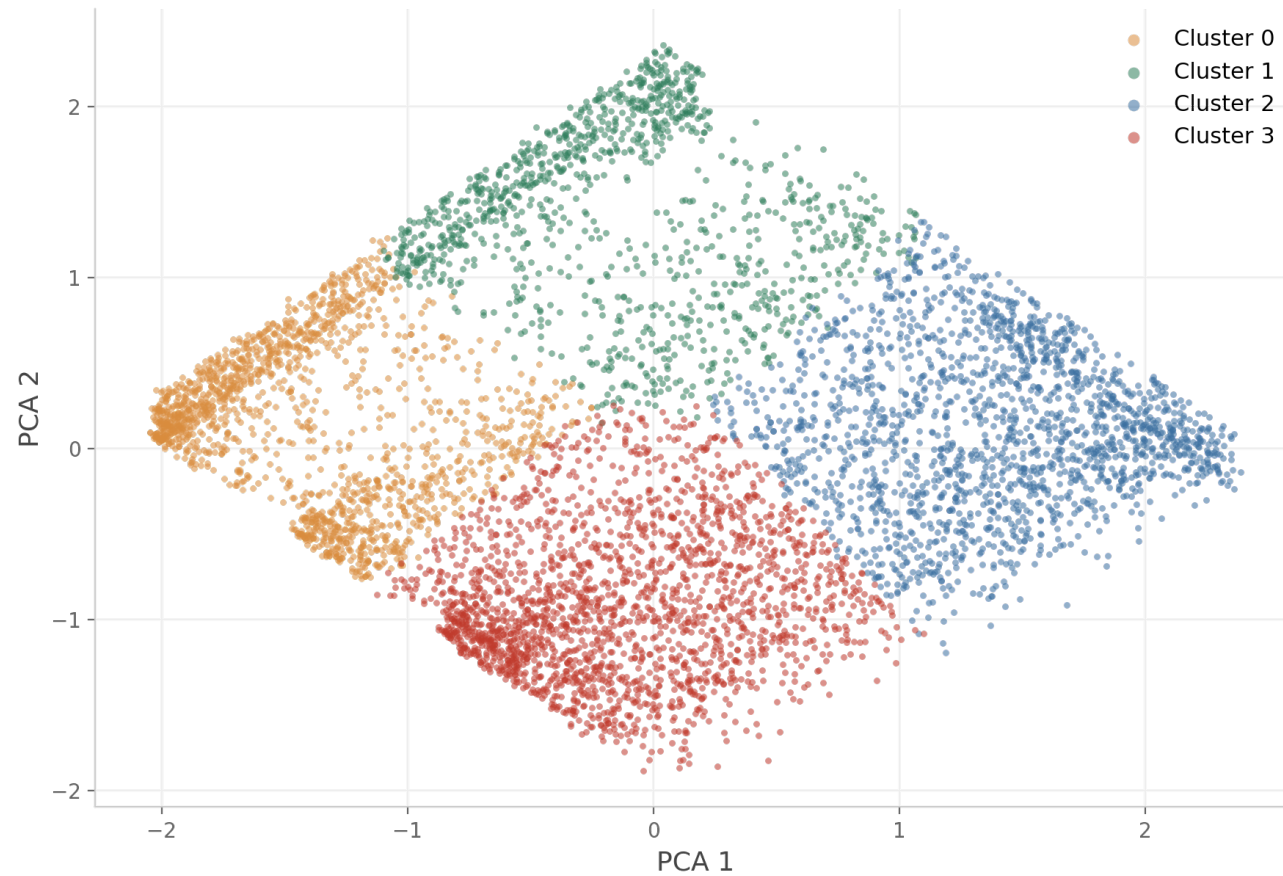
Elbow Method (WCSS) & Silhouette Score across $k = 1-10$



k = 4 selected. Inertia flattens sharply after $k=4$ (elbow bend). Silhouette peaks at $k=2$ (a trivial split) then stabilises ≈ 0.17 from $k=4$ onward — confirming 4 as the best balance of statistical separation and interpretability.

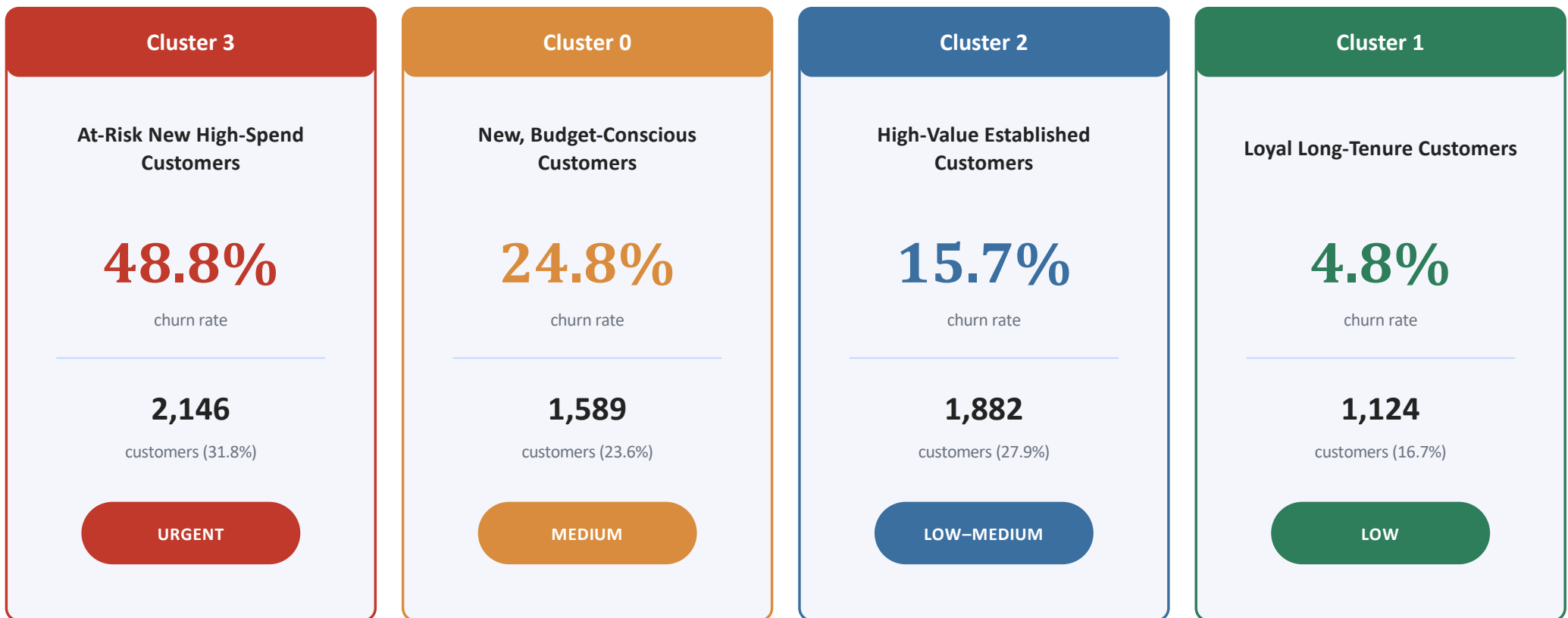
Customer Segments — PCA Projection

All 6,741 customers, reduced to 2 dimensions via PCA (49.8% of variance explained)



Segment Overview

Four behaviourally distinct customer segments emerged



Cluster 3 — “At-Risk New High-Spend Customers”

URGENT

The most critical cluster and the largest segment in the customer base. Short-tenure customers who already pay above-average charges — and carry the highest churn risk by far.

Key Characteristics

- Short tenure — well below the customer-base average
- Above-average monthly charges despite the short relationship
- Highest Senior Citizen share of any cluster — 24.3%
- Lowest share with dependents — 18.1% (mostly individual customers)
- 2,146 customers — 31.8% of total base (largest segment)

Business Insight

Combines the highest churn rate with the largest customer count — driving the majority of lost revenue. Urgent priority for retention.

48.8%

churn rate

2,146

customers (31.8% of base)

PRIORITY

URGENT



Cluster 0 — “New, Budget-Conscious Customers”

Newer customers paying the lowest monthly charges of any group, with short tenure and a fairly even mix of contract types.

Key Characteristics

- Short tenure — below the customer-base average
- Lowest monthly charges of all clusters
- Low Senior Citizen share — 7.2%
- Mixed contracts — about half (50.8%) month-to-month
- 1,589 customers — 23.6% of total base

Business Insight

Price-sensitive and still forming loyalty. Focus on onboarding and incentives to move them onto longer contracts.

24.8%

churn rate

1,589

customers (23.6% of base)

PRIORITY

MEDIUM



Cluster 2 — “High-Value Established Customers”

A stable, established base with the longest average tenure and the highest monthly charges of any cluster.

Key Characteristics

- Longest average tenure of all clusters
- Highest monthly charges of all clusters
- Highest Senior Citizen share — 21.7%
- Nearly universal phone service — 99.7%
- 1,882 customers — 27.9% of total base

Business Insight

A valuable, long-standing segment. Maintain service quality and offer senior-specific perks to protect this revenue base.

15.7%

churn rate

1,882

customers (27.9% of base)

PRIORITY

LOW–MEDIUM



Cluster 1 — “Loyal Long-Tenure Customers”

The most loyal segment by far — long tenure, below-average charges, and the highest share of customers with dependents.

Key Characteristics

- Most loyal group — lowest churn rate of all clusters
- Long average tenure — well above the customer-base average
- Below-average monthly charges
- Highest share with dependents — 44.8%
- 1,124 customers — 16.7% of total base

Business Insight

The most stable, reliable source of long-term revenue. Reward with loyalty programmes and family bundles — requires the least intervention.

4.8%

churn rate

1,124

customers (16.7% of base)

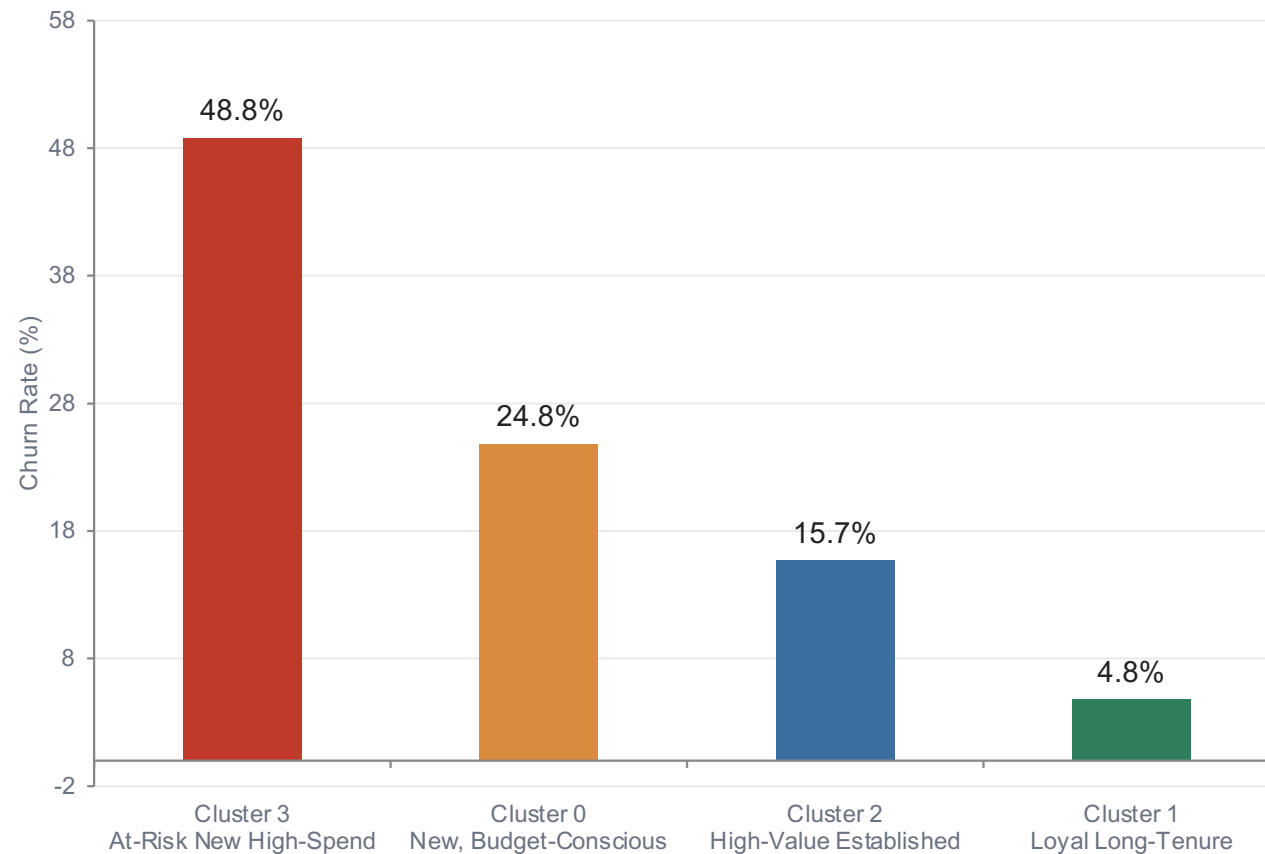
PRIORITY

LOW



Churn Rate by Cluster

The clearest actionable signal in this analysis



Overall average churn: 26.6%

Cluster 3 churns at **10×** the rate of Cluster 1 (48.8% vs 4.8%) — and is also the largest single segment.

That single cluster represents the greatest revenue risk in the portfolio.

Recommendations

Priority actions, ranked by segment urgency

1

Retain Cluster 3 first

Launch proactive retention campaigns for at-risk, new high-spend customers — the top priority.

2

Senior-specific plans

Introduce senior-tailored pricing & support, given the high senior share in Clusters 2 and 3.

3

Strengthen onboarding

Improve early-tenure engagement for Clusters 0 and 3 to build loyalty sooner.

4

Incentivise longer contracts

Offer long-term contract discounts to shift customers off month-to-month plans.

5

Protect loyal segments

Maintain service quality for Clusters 1 and 2; reinforce with family bundle offers for Cluster 1.

Thank You

Questions & discussion welcome

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